

Comprehensive fixed-order analysis of dileptonic top-pair production

Observable definitions and implementation reference

fNLO fixed-order template

30 August 2026

Abstract

This document specifies the comprehensive HwU analysis `analysis_HwU_pp_ttx_dilepton.o` for $pp \rightarrow t\bar{t} + X \rightarrow b\ell^+ \nu_\ell \bar{b}\ell^- \bar{\nu}_\ell + X$ at the LHC. Its 247 histograms cover inclusive truth information, charged leptons, neutrinos, reconstructed W bosons and top quarks, FastJet jet activity, a detector-motivated fiducial region, transverse variables that do not require the individual neutrino momenta, and the complete set of single- and double-spin angular moments in the k, r, n basis. The same IR-safe measurement function is used for additive and multiplicative NLO decay-chain events and for all real-event counterevents.

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1 Scope and intended use

The analysis accepts one final-state positive charged lepton, one negative charged lepton, their flavour-matched neutrino and antineutrino, one b , and one $\bar{b}$. Here $\ell = e, \mu, \tau$; particle identification is by PDG code and status rather than by position in the process definition. The positive-lepton chain is always associated with the top decay and the negative-lepton chain with the antitop decay. Extra final-state light quarks and gluons are allowed. Events that do not have a unique required set of decay products stop with an explicit error instead of silently filling an incorrect observable.

There are three complementary levels:

1. **Inclusive truth** uses the exact neutrino momenta and perfect identification of decay products, with no acceptance cuts. It is useful for theory comparisons, spin-density studies, and understanding decay dynamics, but is not directly measurable.
2. **Fiducial visible** applies lepton, jet, isolation, dilepton-mass, and missing-momentum requirements. It never uses the individual true neutrino momenta. These observables are closest to stable-particle or detector-level measurements, subject to the parton-level limitations in Sec. 10.
3. **Neutrino-blind reconstructed** starts from the fiducial objects and total missing transverse momentum. A deterministic mass-constraint fit supplies two neutrino four-momenta, reconstructed $W^\pm$, top and antitop candidates, and reconstructed spin angles. This is a transparent fixed-order proxy, not an emulation of a particular experimental fit.

All plots are filled with every active HwU weight. Thus the central value, scale variations, PDF variations, and any other weights made available by the run are propagated consistently. Histogram integrals have the normal fixed-order cross-section normalization; no shape normalization is applied by the analysis.

2 Kinematic conventions

Four-vectors are $p = (E, p_x, p_y, p_z)$, with

$$p_T(p) = \sqrt{p_x^2 + p_y^2}, \quad y(p) = \frac{1}{2} \ln \frac{E + p_z}{E - p_z}, \quad (1)$$

$$\eta(p) = \frac{1}{2} \ln \frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}, \quad m(p) = \sqrt{\max(0, p^2)}. \quad (2)$$

The unsigned azimuthal separation lies in $[0, \pi]$, while $\Delta\phi_{\text{signed}}(a, b) = \text{atan2}[\sin(\phi_a - \phi_b), \cos(\phi_a - \phi_b)]$. Angular distance is $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. “Leading” and “trailing” mean largest and smallest p_T , respectively.

The total missing transverse vector is constructed only from visible final-state particles,

$$\vec{p}_T^{\text{miss}} = - \sum_{i \notin \{\nu_e, \nu_\mu, \nu_\tau\}} \vec{p}_{T,i}. \quad (3)$$

Momentum conservation makes it equal to the dineutrino transverse momentum for an ideal event, but the realistic reconstruction never reads either true neutrino momentum separately.

For a visible system v and a massless invisible transverse momentum $\vec{q}_T$,

$$m_T^2(v, \vec{q}_T) = m_v^2 + 2(E_{T,v}|\vec{q}_T| - \vec{p}_{T,v} \cdot \vec{q}_T), \quad E_{T,v} = \sqrt{m_v^2 + p_T^2(v)}. \quad (4)$$

The dilepton cluster transverse mass treats the two neutrinos as one massless invisible system. The stransverse mass is

$$m_{T2}(v_1, v_2) = \min_{\vec{q}_{1T} + \vec{q}_{2T} = \vec{p}_T^{\text{miss}}} \max\{m_T(v_1, \vec{q}_{1T}), m_T(v_2, \vec{q}_{2T})\}. \quad (5)$$

The code evaluates this two-dimensional minimum with a coarse grid followed by a convergent pattern search. For $m_{T2}^{b\ell}$, both lepton-b-jet pairings are minimized and the smaller result is retained.

3 Truth resonances and IR-safe jets

The exact W candidates are $p_{W^+} = p_{\ell^+} + p_\nu$ and $p_{W^-} = p_{\ell^-} + p_{\bar{\nu}}$. At Born level the top candidates are $p_{W^+} + p_b$ and $p_{W^-} + p_{\bar{b}}$. With real radiation, every non-bottom final-state QCD parton is assigned to production, the top decay, or the antitop decay. Of the $3^{N_{\text{rad}}}$ assignments, the analysis picks the one minimizing

$$\chi_{\text{truth}}^2 = \left(\frac{m_{t,\text{cand}} - m_t^{\text{ref}}}{5 \text{ GeV}} \right)^2 + \left(\frac{m_{\bar{t},\text{cand}} - m_{\bar{t}}^{\text{ref}}}{5 \text{ GeV}} \right)^2. \quad (6)$$

This makes the reconstruction independent of event-record ordering and keeps decay radiation with the resonance that it reconstructs. Soft and collinear limits are unchanged except on measure-zero assignment boundaries. The reference masses are read from the model parameters.

All final-state quarks with $|\text{PDG}| \leq 5$ and all gluons are passed to FastJet. Jets use the anti- k_T algorithm with $R = 0.4$; unless stated otherwise, counted jets have $p_T > 25 \text{ GeV}$ and $|\eta| < 4.5$. A jet is ideally b-tagged when it contains the unique final-state b or $\bar{b}$ parton. If both are clustered into one jet, inclusive jet plots remain well-defined but the two-distinct-b-jet fiducial requirement fails. Additional jets are all accepted jets other than the two b-tagged jet indices.

Define

$$H_T^j = \sum_j p_T(j), \quad H_T^{\text{extra}} = \sum_{j \notin b, \bar{b}} p_T(j), \quad (7)$$

$$H_T^{\text{vis}} = p_T(\ell^+) + p_T(\ell^-) + H_T^j, \quad S_T = H_T^{\text{vis}} + p_T^{\text{miss}}, \quad (8)$$

$$p_{\text{vis}} = p_{\ell^+} + p_{\ell^-} + p_{b\text{-jet}} + p_{\bar{b}\text{-jet}}. \quad (9)$$

The two jet-gap histograms are cumulative cross sections, not differential spectra: their bin centred at Q receives an event when the leading additional-jet $p_T < Q$, or when $H_T^{\text{extra}} < Q$, respectively. Threshold centres run from 15 to 205 GeV in 10 GeV steps.

4 Fiducial region and cutflow

The baseline fiducial selection is deliberately simple and reproducible:

$$\begin{aligned} p_T(\ell_1) &> 25 \text{ GeV}, & p_T(\ell_2) &> 20 \text{ GeV}, & |\eta(\ell)| &< 2.5, \\ m_{\ell\ell} &> 20 \text{ GeV}, & p_T(b\text{-jet}) &> 30 \text{ GeV}, & |\eta(b\text{-jet})| &< 2.5, \\ \min_{\ell,j} \Delta R(\ell, j) &> 0.4, & p_T^{\text{miss}} &> 20 \text{ GeV}. \end{aligned}$$

Two distinct b-tagged jets are required. The isolation minimum includes all jets with $p_T > 25 \text{ GeV}$ and $|\eta| < 4.5$. An additional same-flavour signal-region stage requires $|m_{\ell\ell} - 91.1876 \text{ GeV}| > 15 \text{ GeV}$ and $p_T^{\text{miss}} > 40 \text{ GeV}$; different-flavour events pass this stage without the extra requirements.

The eight cutflow bin centres mean:

Bin	Requirement accumulated through this bin
1	all valid dileptonic events
2	lepton p_T and $ \eta $
3	dilepton mass
4	two distinct accepted b-tagged jets
5	lepton-jet separation
6	baseline missing transverse momentum (the fiducial sample)
7	same-flavour Z -veto and tightened p_T^{miss} , when applicable
8	successful neutrino reconstruction after bin 7

The visible $b\ell$ pairing chooses between $\{\ell^+b, \ell^-\bar{b}\}$ and $\{\ell^+\bar{b}, \ell^-b\}$ by selecting the pairing with the smaller maximum pair mass. The minimum, maximum, and average masses in the chosen pairing are plotted. This definition uses no neutrino information.

5 Neutrino-blind reconstruction

For a trial transverse split $\vec{q}_T$ and $\vec{p}_T^{\text{miss}} - \vec{q}_T$, the two equations $(p_\ell + p_\nu)^2 = (m_W^{\text{ref}})^2$ give up to two longitudinal roots per side. When a discriminant is negative, its real part is used. Both roots and both b-jet/lepton pairings are tested. The selected solution minimizes

$$\chi_{\text{reco}}^2 = \sum_{q=t,\bar{t}} \left(\frac{m_q - m_t^{\text{ref}}}{15 \text{ GeV}} \right)^2 + \sum_{V=W^+,W^-} \left(\frac{m_V - m_W^{\text{ref}}}{10 \text{ GeV}} \right)^2 + \left(\frac{m_t - m_{\bar{t}}}{30 \text{ GeV}} \right)^2. \quad (10)$$

The transverse split is initialized on a 5×5 grid around half of $\vec{p}_T^{\text{miss}}$, followed by up to twelve eight-direction pattern-search refinements. The procedure is deterministic and uses only the two charged leptons, two b-tagged jets, total $\vec{p}_T^{\text{miss}}$, and the model pole masses. Consequently reconstructed truth-comparison plots are validation diagnostics; all other reconstructed plots are in principle obtainable without observing either neutrino separately.

6 Spin-correlation observables

Spin axes are constructed in the $t\bar{t}$ rest frame. Let $\hat{p}$ be the boosted positive-beam direction, $\hat{k}$ the top direction, and $c = \hat{p} \cdot \hat{k}$. The beam-basis triad is

$$\hat{r} = \text{sign}(c) \frac{\hat{p} - c\hat{k}}{\sqrt{1-c^2}}, \quad \hat{n} = \text{sign}(c) \frac{\hat{p} \times \hat{k}}{\sqrt{1-c^2}}, \quad \hat{k} = \frac{\vec{p}_t^{t\bar{t}}}{|\vec{p}_t^{t\bar{t}}|}. \quad (11)$$

The sign factors make the axes appropriate for a symmetric pp initial state. A deterministic orthogonal completion is used at the exceptional forward points. The charged leptons are boosted first into the $t\bar{t}$ frame and then into their respective parent rest frames. With $a, b \in \{k, r, n\}$,

$$c_a^+ = \hat{\ell}_+ \cdot \hat{a}, \quad c_b^- = \hat{\ell}_- \cdot \hat{b}, \quad q_{ab} = c_a^+ c_b^-. \quad (12)$$

The six one-spin angles and all nine products are histogrammed. This is the complete angular information needed for the single-spin polarizations and the 3×3 spin-correlation matrix. In the commonly used convention

$$\frac{1}{\sigma} \frac{d^2\sigma}{dc_a^+ dc_b^-} = \frac{1}{4} [1 + B_a^+ c_a^+ + B_b^- c_b^- - C_{ab} c_a^+ c_b^-], \quad (13)$$

the moments obey $B_a^\pm = 3\langle c_a^\pm \rangle$ and $C_{ab} = -9\langle q_{ab} \rangle$. Stating this convention matters: the histograms themselves are raw, convention-independent angular moments.

The symmetric and antisymmetric off-diagonal combinations expose $C_{ab} + C_{ba}$ and $C_{ab} - C_{ba}$. The event-level diagonal quantity is $d_{\text{diag}} = (q_{kk} + q_{rr} + q_{nn})/3$; with the convention above, $D \equiv \text{tr } C/3 = -9\langle d_{\text{diag}} \rangle$. The lepton opening cosine, laboratory $\Delta\phi_{\ell\ell}$, signed $\sin \Delta\phi_{\ell\ell}$, and $(\hat{\ell}_+ \times \hat{\ell}_-) \cdot \hat{k}$ provide compact spin and CP-sensitive projections. The complete set is filled once with exact truth tops (histograms 191–216) and once with reconstructed tops (217–242).

For the W -helicity angles, the charged lepton is evaluated in its W rest frame against the direction opposite to the top-decay recoil in that frame. This exact definition is used for both charges and avoids ambiguity about the anti-top decay-product ordering.

7 Complete histogram catalogue

The integer in the first column is the booking order in **MADatNLO.HwU**. Every name shown below is followed by the HwU tag **!T@NLO**. “Truth” means no acceptance selection; “fiducial” means the baseline through cutflow bin 6; “reconstructed” additionally requires a valid mass-constraint solution.

7.1 Rates and truth top-pair kinematics (1–28)

ID	HwU name	Definition and purpose
1	inclusive rate	One-bin total valid-process rate.
2	fiducial cutflow	Eight cumulative stages defined in Sec. 4.
3–4	truth top p_T ; truth antitop p_T	$p_T(t)$ and $p_T(\bar{t})$.

ID	HwU name	Definition and purpose
5–6	truth leading top pT; truth trailing top pT	Maximum and minimum of the two top transverse momenta.
7–8	truth top rapidity; truth antitop rapidity	y_t and $y_{\bar{t}}$.
9	truth ttbar mass	$m_{t\bar{t}}$, central for threshold and boosted-tail studies.
10	truth ttbar pT	Recoil $p_T(t\bar{t})$, zero at production Born level but populated by radiation.
11–12	truth ttbar rapidity; truth abs ttbar rapidity	$y_{t\bar{t}}$ and $ y_{t\bar{t}} $.
13–14	truth delta rapidity top antitop; truth abs delta rapidity top antitop	$y_t - y_{\bar{t}}$ and its absolute value.
15	truth delta abs rapidity top antitop	$ y_t - y_{\bar{t}} $, a charge-asymmetry observable.
16	truth ttbar boost rapidity	$(y_t + y_{\bar{t}})/2$.
17	truth ttbar star rapidity	$(y_t - y_{\bar{t}})/2$.
18	truth top scattering cosine	Cosine between the top and positive beam in the $t\bar{t}$ frame.
19	truth top speed in ttbar frame	$ \vec{p}_t^{t\bar{t}} /E_t^{t\bar{t}}$.
20	truth rho two mt over mtt	$\rho = 2m_t^{\text{ref}}/m_{t\bar{t}}$, emphasizing threshold.
21–22	truth delta phi top antitop; truth delta R top antitop	Top-pair azimuthal and η - ϕ separations.
23	truth scalar top pT sum	$p_T(t) + p_T(\bar{t})$.
24	truth top pT asymmetry	$[p_T(t) - p_T(\bar{t})]/[p_T(t) + p_T(\bar{t})]$.
25	truth ttbar transverse centrality	$2[p_T(t) + p_T(\bar{t})]/m_{t\bar{t}}$.
26	truth top energy in ttbar frame	$E_t^{t\bar{t}}$.
27	truth chi exp abs delta y	$\chi = \exp y_t - y_{\bar{t}} $.
28	truth ttbar longitudinal boost	$ p_{z,t\bar{t}} /E_{t\bar{t}}$.

7.2 Truth charged-lepton observables (29–52)

ID	HwU name	Definition and purpose
29–30	truth positive lepton pT; truth negative lepton pT	Charge-separated lepton transverse momenta.
31–32	truth leading lepton pT; truth trailing lepton pT	Ordered lepton transverse momenta.
33–34	truth positive lepton eta; truth negative lepton eta	Charge-separated pseudorapidities.
35–36	truth positive lepton rapidity; truth negative lepton rapidity	Charge-separated true rapidities.
37–40	truth dilepton mass; truth dilepton pT; truth dilepton rapidity; truth abs dilepton rapidity	$m_{\ell\ell}$, $p_T(\ell\ell)$, $y_{\ell\ell}$, and $ y_{\ell\ell} $.
41	truth dilepton delta phi	Unsigned $\Delta\phi(\ell^+, \ell^-)$, a robust laboratory spin observable.
42–43	truth dilepton delta eta; truth abs dilepton delta eta	$\eta_+ - \eta_-$ and its absolute value.
44	truth dilepton delta R	$\Delta R(\ell^+, \ell^-)$.
45	truth lepton opening cosine lab	$\hat{\ell}_+^{\text{lab}} \cdot \hat{\ell}_-^{\text{lab}}$.
46	truth dilepton cos theta star	$\tanh[(\eta_+ - \eta_-)/2]$, a longitudinally boost-invariant scattering proxy.
47	truth lepton pT ratio	$p_T(\ell_2)/p_T(\ell_1)$.
48	truth scalar lepton pT sum	$H_T^\ell = p_T(\ell^+) + p_T(\ell^-)$.
49	truth delta abs lepton eta	$ \eta_+ - \eta_- $, sensitive to charge-dependent centrality.
50	truth dilepton acoplanarity	$\pi - \Delta\phi_{\ell\ell}$.
51	truth signed dilepton delta phi	Wrapped $\phi_+ - \phi_- \in [-\pi, \pi]$.
52	truth dilepton mass over lepton HT	$m_{\ell\ell}/H_T^\ell$.

7.3 Truth W-boson, decay, and neutrino observables (53–91)

ID	HwU name	Definition and purpose
53–55	truth Wplus pT; truth Wplus rapidity; truth Wplus mass	p_T , y , and invariant mass of $p_{\ell^+} + p_\nu$.
56–58	truth Wminus pT; truth Wminus rapidity; truth Wminus mass	Corresponding quantities for $p_{\ell^-} + p_{\bar{\nu}}$.
59–61	truth WW mass; truth WW pT; truth WW rapidity	Invariant mass, transverse momentum, and rapidity of the diboson system.
62–63	truth WW delta phi; truth WW delta R	Angular separations of the two truth W candidates.
64–65	truth positive lepton b mass; truth negative lepton bbar mass	Charge-correct $m(\ell^+ b\text{-jet})$ and $m(\ell^- \bar{b}\text{-jet})$.
66–68	truth minimum correct lb mass; truth maximum correct lb mass; truth average correct lb mass	Minimum, maximum, and mean of the two charge-correct masses.
69–70	truth positive lepton energy top frame; truth negative lepton energy top frame	Charged-lepton energies in the associated exact parent rest frames.
71–72	truth top recoil energy top frame; truth antitop recoil energy top frame	Energy of $p_t - p_W$ in each parent frame; includes assigned decay radiation.
73–74	truth Wplus helicity cosine; truth Wminus helicity cosine	Charged-lepton helicity angles defined in Sec. 6.
75–76	truth Wplus energy top frame; truth Wminus energy top frame	W -candidate energy in its parent rest frame.
77–79	truth neutrino pT; truth neutrino eta; truth neutrino rapidity	Exact top-decay neutrino kinematics; theory-only.
80–82	truth antineutrino pT; truth antineutrino eta; truth antineutrino rapidity	Exact antitop-decay antineutrino kinematics; theory-only.
83–85	truth dineutrino pT; truth dineutrino mass; truth dineutrino rapidity	Kinematics of $p_\nu + p_{\bar{\nu}}$.
86–87	truth dineutrino delta phi; truth dineutrino delta R	Separation of the two exact neutrinos; theory-only.
88	visible missing pT	Magnitude of missing momentum inferred from the visible final state.
89–90	minimum delta phi missing pT lepton; maximum delta phi missing pT lepton	Smaller and larger of the two lepton–missing-vector azimuthal separations.
91	dilepton missing transverse mass	Cluster transverse mass of $p_{\ell\ell}$ and total missing momentum.

7.4 FastJet and event-activity observables (92–121)

ID	HwU name	Definition and purpose
92–93	truth b jet pT; truth bbar jet pT	Ideal flavour-labelled b-jet transverse momenta.
94–95	truth leading b jet pT; truth trailing b jet pT	Ordered b-jet transverse momenta.
96–97	truth b jet eta; truth bbar jet eta	Ideal flavour-labelled b-jet pseudorapidities.
98–101	truth bb jet mass; truth bb jet pT; truth bb jet delta phi; truth bb jet delta R	Mass, recoil, and separation of the two b-jet system.
102–103	jet multiplicity pT25; jet multiplicity pT40	Numbers of jets with $ \eta < 4.5$ above 25 and 40 GeV.
104–107	leading jet pT; leading jet eta; subleading jet pT; subleading jet eta	Kinematics of the first two accepted jets, ordered by FastJet p_T .
108	additional jet multiplicity pT25	Accepted jets excluding the b- and $\bar{b}$ -tagged jet indices.
109–110	leading additional jet pT; leading additional jet rapidity	Hardest additional-jet kinematics when one is present.
111	additional jet HT	H_T^{extra} .
112	jet HT pT25	H_T^j .

ID	HwU name	Definition and purpose
113	visible HT leptons and jets	H_T^{vis} .
114	visible ST including missing pT	S_T .
115–117	visible dilepton bb mass; visible dilepton bb pT; visible dilepton bb rapidity	Kinematics of p_{vis} .
118	additional jet gap Q_0	Cumulative rate with leading additional-jet $p_T < Q_0$.
119	additional jet gap Q_{sum}	Cumulative rate with $H_T^{\text{extra}} < Q_{\text{sum}}$.
120–121	minimum lepton b jet delta R; maximum lepton b jet delta R	Minimum and maximum among all four charged-lepton/b-jet pair distances.

7.5 Fiducial visible observables (122–159)

ID	HwU name	Definition and purpose
122	fiducial rate	One-bin baseline fiducial cross section.
123–124	fiducial leading lepton pT; fiducial trailing lepton pT	Ordered selected-lepton transverse momenta.
125–126	fiducial positive lepton eta; fiducial negative lepton eta	Charge-separated selected-lepton pseudorapidities.
127–129	fiducial dilepton mass; fiducial dilepton pT; fiducial dilepton rapidity	Selected dilepton-system kinematics.
130–132	fiducial dilepton delta phi; fiducial abs dilepton delta eta; fiducial dilepton delta R	Selected dilepton angular separations.
133	fiducial missing pT	Visible-state p_T^{miss} after the baseline cuts.
134	fiducial dilepton missing transverse mass	Dilepton cluster transverse mass.
135	fiducial min delta phi missing lepton	Smaller lepton–missing-vector azimuthal separation.
136–139	fiducial leading b jet pT; fiducial trailing b jet pT; fiducial abs leading b jet eta; fiducial abs trailing b jet eta	Ordered accepted b-jet kinematics.
140–142	fiducial bb jet mass; fiducial bb jet pT; fiducial bb jet delta R	Selected b-jet-pair kinematics.
143–145	fiducial visible dilepton bb mass; fiducial visible dilepton bb pT; fiducial visible dilepton bb rapidity	Kinematics of the selected visible $\ell\ell b\bar{b}$ system.
146–148	fiducial jet HT; fiducial visible HT; fiducial ST	Selected H_T^j , H_T^{vis} , and S_T .
149–150	fiducial jet multiplicity; fiducial additional jet multiplicity	Selected total and non-b jet counts.
151–153	fiducial minimum paired lb mass; fiducial maximum paired lb mass; fiducial average paired lb mass	Three masses from the neutrino-blind minimax pairing.
154	fiducial mT2 dilepton	$m_{T2}(\ell^+, \ell^-)$ with massless invisible daughters.
155	fiducial mT2 lb systems	Minimum m_{T2} of the two possible $b\ell$ pairings.
156	fiducial delta abs lepton eta	$ \eta_+ - \eta_- $.
157	fiducial lepton pT ratio	Trailing-to-leading lepton p_T ratio.
158	fiducial b jet pT ratio	Trailing-to-leading b-jet p_T ratio.
159	fiducial lepton opening cosine lab	Laboratory three-dimensional charged-lepton opening cosine.

7.6 Neutrino-blind reconstructed observables (160–190)

ID	HwU name	Definition and purpose
160	reconstructed rate	One-bin rate for successful reconstruction in the baseline fiducial region.

ID	HwU name	Definition and purpose
161–163	reconstructed Wplus pT; reconstructed Wplus rapidity; reconstructed Wplus mass	Fitted W^+ candidate kinematics.
164–166	reconstructed Wminus pT; reconstructed Wminus rapidity; reconstructed Wminus mass	Fitted W^- candidate kinematics.
167–169	reconstructed top pT; reconstructed top rapidity; reconstructed top mass	Fitted top kinematics.
170–172	reconstructed antitop pT; reconstructed antitop rapidity; reconstructed antitop mass	Fitted antitop kinematics.
173–175	reconstructed ttbar mass; reconstructed ttbar pT; reconstructed ttbar rapidity	Fitted top-pair-system kinematics.
176	reconstructed delta top rapidity	$y_t^{\text{reco}} - y_{\bar{t}}^{\text{reco}}$.
177	reconstructed delta abs top rapidity	$ y_t^{\text{reco}} - y_{\bar{t}}^{\text{reco}} $.
178	reconstructed top scattering cosine	Reconstructed top/beam angle in the reconstructed $t\bar{t}$ frame.
179	reconstructed top speed ttbar frame	Reconstructed $ \vec{p}_t /E_t$ in that frame.
180–181	reconstructed neutrino pz; reconstructed antineutrino pz	Longitudinal roots selected by the global constraint score.
182–183	reconstructed dineutrino mass; reconstructed dineutrino pz	Mass and longitudinal momentum of the fitted neutrino pair.
184	reconstruction constraint score	Selected χ_{reco}^2 , clipped only at the histogram edge.
185	reconstructed top mass difference	$m_t^{\text{reco}} - m_{\bar{t}}^{\text{reco}}$.
186	reconstructed b lepton pairing	Bin 1 is charge-correct ($\ell^+b, \ell^-\bar{b}$); bin 2 is crossed.
187	reco truth average top pT ratio	$[p_T(t)_{\text{reco}} + p_T(\bar{t})_{\text{reco}}]/[p_T(t)_{\text{truth}} + p_T(\bar{t})_{\text{truth}}]$; validation only.
188	reco truth ttbar mass residual	$m_{t\bar{t}}^{\text{reco}} - m_{t\bar{t}}^{\text{truth}}$; validation only.
189–190	reconstructed Wplus helicity cosine; reconstructed Wminus helicity cosine	Helicity angles from fitted W and top candidates.

7.7 Truth and reconstructed spin sets (191–242)

The following 26-entry template occurs twice. In the table, $s = 191$ for the truth set and $s = 217$ for the reconstructed set; the ID is $s + i - 1$. The exact HwU prefix is **truth spin** or **reco spin**.

i (IDs)	Suffix	Definition
1–3 (191–193; 217–219)	cos plus k/r/n	c_k^+, c_r^+, c_n^+ : positive-lepton one-spin analysers.
4–6 (194–196; 220–222)	cos minus k/r/n	c_k^-, c_r^-, c_n^- : negative-lepton one-spin analysers.
7–15 (197–205; 223–231)	product kk, kr, kn, rk, rr, rn, nk, nr, nn	All $q_{ab} = c_a^+ c_b^-$, in row-major $a, b = k, r, n$ order; the full correlation matrix.
16 (206; 232)	lepton opening cosine	$\hat{\ell}_+ \cdot \hat{\ell}_-$ in their respective parent rest frames, using common k, r, n coordinates.
17 (207; 233)	lab dilepton delta phi	Unsigned laboratory $\Delta\phi_{\ell\ell}$.

i (IDs)	Suffix	Definition
18–19 (208– 209; 234– 235)	symmetric kr rk; antisymmetric kr rk	$q_{kr} + q_{rk}$ and $q_{kr} - q_{rk}$.
20–21 (210– 211; 236– 237)	symmetric kn nk; antisymmetric kn nk	$q_{kn} + q_{nk}$ and $q_{kn} - q_{nk}$.
22–23 (212– 213; 238– 239)	symmetric rn nr; antisymmetric rn nr	$q_{rn} + q_{nr}$ and $q_{rn} - q_{nr}$.
24 (214; 240)	lepton triple product	$(\hat{\ell}_+ \times \hat{\ell}_-) \cdot \hat{k}$, a signed naive-T-odd projection.
25 (215; 241)	diagonal D estimator	Event variable $(q_{kk} + q_{rr} + q_{nn})/3$.
26 (216; 242)	signed sine delta phi	$\sin[\Delta\phi_{\text{signed}}(\ell^+, \ell^-)]$.

7.8 Reconstruction diagnostics (243–247)

ID	HwU name	Definition and purpose
243	diagnostic truth top assignment score	χ_{truth}^2 of the selected radiation assignment.
244	diagnostic assigned radiation count	Number of light QCD partons assigned to either top decay rather than production.
245	diagnostic truth top mass residual	$\max(m_t - m_t^{\text{ref}} , m_{\bar{t}} - m_{\bar{t}}^{\text{ref}})$.
246	diagnostic reco W mass residual	$\max(m_{W^+}^{\text{reco}} - m_W^{\text{ref}} , m_{W^-}^{\text{reco}} - m_W^{\text{ref}})$.
247	diagnostic reconstruction rate	One-bin successful-reconstruction rate; compare with the fiducial rate.

8 Fixed-order and IR-safety considerations

The analysis is called on real events and on their soft, collinear, and soft-collinear projections. It deliberately does not branch on the internal body label. Anti- k_T jets, inclusive missing momentum, the deterministic minimizations, and all measurement cuts are applied identically to each event/counterevent. Therefore unresolved momenta that are indistinguishable to an IR-safe observable remain indistinguishable to the analysis. Sharp cut boundaries and discrete assignment boundaries can differ only on sets of zero phase-space measure, as in an ordinary fixed-order analysis.

The truth-top assignment is resonance-aware but not an event-history query. This matters for the multiplicative path, whose combined event can contain different real/Born momentum choices for different production and decay blocks. The analysis reconstructs solely from the materialized external state, so additive and multiplicative results use the same observable definition.

Jet multiplicities and jet vetoes are meaningful fixed-order observables, but large logarithms can make the exclusive zero-additional-jet region sensitive to resummation. Similarly, tails near kinematic endpoints can carry large scale dependence. HwU scale/PDF variations should be inspected rather than interpreting only a central curve.

9 Running the analysis

Select the module in `Cards/F0_analyse_card.dat`:

```
FO_ANALYSIS_FORMAT = HwU
```

```
FO_ANALYSE          = analysis_HwU_pp_ttx_dilepton.o
```

The companion ABI bridge is selected automatically. FastJet is already part of the fixed-order template link, so no extra FastJet entry is needed in `FO_EXTRALIBS`. The module works for both `NLO_DECAY_COMBINATION ADDITIVE` and `NLO_DECAY_COMBINATION MULTIPLICATIVE`. The output is `MADatNLO.HwU`; standard HwU post-processing may be used for uncertainty bands, normalized distributions, ratios, and publication plots.

For a direct-decay process, a representative syntax is

```
generate p p > t t~ [QCD], \
(t > w+ b QED=1 [QCD], w+ > e+ ve), \
(t~ > w- b~ QED=1 [QCD], w- > mu- vm~)
```

The lepton flavour and the order of decay products in the process command do not affect object mapping.

10 Limitations and interpretation

- Jets are parton-level FastJet jets. There is no shower, hadronization, pile-up, detector response, lepton dressing, or b-tagging efficiency/mistag model. Ideal constituent flavour labels are used only to identify the two b jets.
- The fiducial cuts are a generic LHC-like reference selection, not the exact selection of a specific ATLAS or CMS publication. A comparison to data must reproduce that measurement’s object definitions, flavour channels, dressing, jet acceptance, and unfolding level.
- The neutrino reconstruction is a fast deterministic constraint fit. Experimental methods can include transfer functions, multiple detector hypotheses, likelihood weights, and resolution smearing. Truth-residual plots 187–188 are not observable in data.
- Exact neutrino, exact W , truth-top, and truth-spin plots assume perfect knowledge and are theory-only. Fiducial lepton, jet, missing- momentum, m_{T2} , and visible-system plots require much less reconstruction and are the preferred measurement-facing set.
- The analysis expects exactly one final charged lepton of each sign and a unique flavour-matched neutrino pair. It does not unfold secondary τ decays or additional electroweak lepton pairs.

11 Primary references

The complete spin-density-matrix organization and the k, r, n basis follow the framework of Bernreuther, Heisler, and Si, arXiv:1508.05271. Experimental use of the full set of 15 independent polarization and correlation coefficients is illustrated by ATLAS, arXiv:1612.07004, and comprehensive dilepton spin measurements by CMS, arXiv:1907.03729.

Representative LHC differential measurements motivating the top, lepton, dilepton, b-jet, and additional-jet catalogue include CMS arXiv:1708.07638 and arXiv:1811.06625, and ATLAS arXiv:1709.09407, arXiv:1606.09490, and arXiv:2303.15340. Dineutrino and missing-momentum-sensitive observables are discussed in the recent CMS study arXiv:2510.00160; the visible $m_{b\ell}$ endpoint is used, for example, in the ATLAS top-mass analysis arXiv:1503.05427.

Jet finding uses FastJet arXiv:1111.6097 and the anti- k_T algorithm of Cacciari, Salam, and Soyez, arXiv:0802.1189.